

Part 5, Container theorem

Lecture 15 Kleitman-Winston graph container thm.

Thm 1.1 (Kleitman-Winston 80s) Let G be an N -vx graph.

Let $R, q \in \mathbb{N}$, $0 < \beta < 1$ s.t.

$$e^{-\beta q} \cdot N \leq R, \quad \dots \dots \dots (1)$$

Suppose $\forall S \subseteq V(G)$ w/ $|S| \geq R$, local denseness

$$e_G(S) \geq \beta \binom{|S|}{2} \quad \dots \dots \dots (2)$$

then

$\Rightarrow \exists$ a collection of (containers) $C_i \subseteq V(G)$, $1 \leq i \leq \binom{N}{q}$,

s.t.

- $|C_i| \leq R + q$;
- $\forall$ indep set I in G , $\exists$ some i s.t. $I \subseteq C_i$.

In particular, # indep sets of size $m \geq q$ is at most

$$\binom{N}{q} \binom{R+q}{m-q}.$$

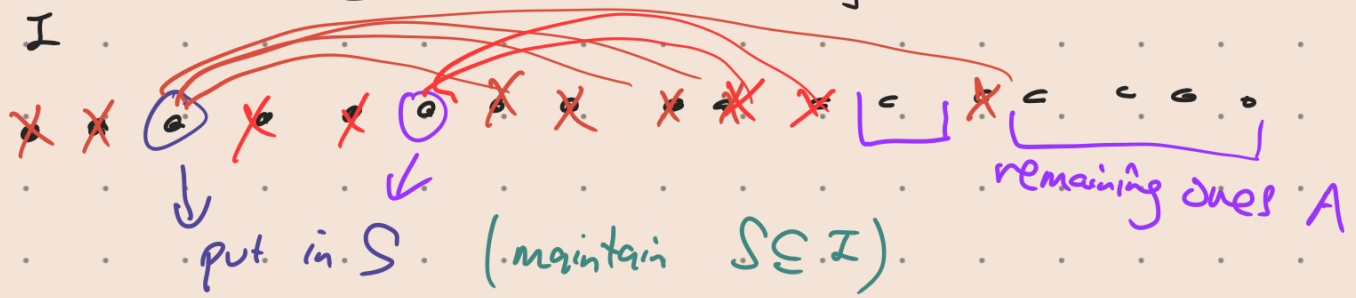
Roughly speaking, it states that provided that G is locally dense, then $\exists$ a 'small' collection of containers for indep sets in G .

Idea: locally dense $\Rightarrow$ group indep sets up using
small fingerprints (the set S later in the pf)

Def Given G , $A \subseteq V(G)$, the max-deg ordering of A is $(v_1, v_2, \dots, v_{|A|})$ s.t.

for each $1 \leq i \leq |A|$, v_i is a maximum-degree vertex in $G[A \setminus \{v_1, \dots, v_{i-1}\}]$.

Given I



- $I \subseteq S \cup A = \text{Container}$
- $A = A(S)$
- Shall see - fingerprint S small
- A not large (locally dense)

Pf : Let I be an indep set in G . Run the following alg.

Input : G, I

Initially: $A = V(G)$ (active set)

$S = \emptyset$ (selected ones)
(or finger print of I)

For the i^{th} iteration, $i = 1, \dots, 2$, do

Step 1 : Select the next I -vx.

- Take the max-deg ordering of A , say $(v_1, \dots, v_{|A|})$.
- Let v_{t_i} be the 1st (min. index) vx lying in I . Move v_{t_i} from A to S .

Step 2. Update A .

Delete $\{v_1, \dots, v_{t_i-1}\} \cup (N_G(v_{t_i}) \cap A)$ from A .

Output: $S = (v_{t_1}, v_{t_2}, \dots, v_{t_q})$ and A .

Claim The final A is not too large

$$|A| \leq R.$$

Assuming the claim,

- $S \subseteq I$ and $I \setminus S \subseteq A$
- the final $A = A(S)$ depends only on S , not on I !

$$\Rightarrow C = S \cup A \supseteq I \quad C = \bigcup_{S \subseteq I} A(S)$$

$$|C| \leq |S| + |A| = q + R.$$

Also # containers $C =$ # fingerprints $S \leq \binom{N}{q}$.

$$\# \text{ indep sets of size } m \leq \binom{N}{q} \cdot \binom{R}{m-q}.$$

Pf of Claim: For each $i \in [q]$, let A_i be the active set prior to i th-iteration.

(so $A_1 = V(G)$, $A_{q+1} = \text{final } A$)

Suppose to the contrary that the final $|A| > R$.

Fix arbitrary $i \in [q]$, let $A' = A_i \setminus \{v_1, \dots, v_{t_i-1}\}$

Then $|A'| \geq |A| > R$.

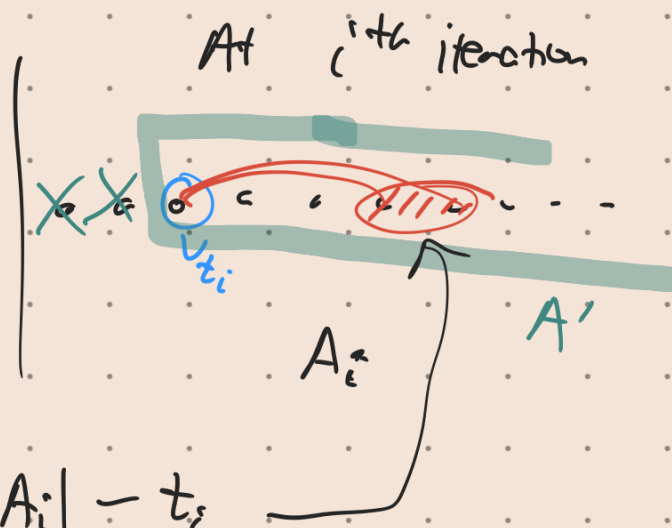
locally dense (2) $\Rightarrow e_G(A') \geq \beta \binom{|A'|}{2}$

$\Rightarrow U_{t_i}$ has deg at least

$$\frac{2e_G(A')}{|A'|} = \beta(|A'| - 1)$$

$$= \beta(|A_i| - t_i + 1) - 1$$

$$= \beta(|A_i| - t_i) \geq \underline{\beta|A_i| - t_i}$$



So # u.s. deleted $(A_i \rightarrow A_{i+1})$

$$\geq t_i + \beta|A_i| - t_i = \beta|A_i|$$

$$\Rightarrow |A_{i+1}| \leq (1 - \beta)|A_i|$$

$$\Rightarrow |A| \leq (1 - \beta)^2 N \leq e^{-\beta^2} \cdot N \leq R \quad \square$$

Application to EKR counting

Recall k -unif hyp. $\mathcal{H}$ on $[n]$ is intersecting if

$$\forall A, B \in \mathcal{H}, A \cap B \neq \emptyset$$

$$\stackrel{\text{EKR}}{\Rightarrow} n \geq 2k, \quad e(\mathcal{H}) \leq \binom{n-1}{k-1}$$

Thm [BDDLS15] $\forall n \geq 2k+1$, # int. k -unif hyp. is

$$2 \binom{n-1}{k-1}$$

• Lower bdd:  $\binom{n-1}{k-1} \Rightarrow \text{subhyp.} \geq 2^{\binom{n-1}{k-1}}$

• Upp bdd $\left\{ \begin{array}{l} \cdot \text{Expander-mixing lem} \Rightarrow \text{Supersaturation} \\ \cdot \text{graph container,} \end{array} \right. \text{local denseness (2)}$

Reduction to counting indep. sets in Kneser graphs

Def.: The Kneser graph $KG(n, k) = (V, E)$

- $V = \binom{[n]}{k}$

- $E: A, B \in \binom{[n]}{k} \quad A \sim B \Leftrightarrow A \cap B = \emptyset$

Intersecting
k-unif hyp.



indep sets
in $KG(n, k)$

$A_1, \dots, A_t$ indep set
 $A_i \not\sim A_j \Leftrightarrow A_i \cap A_j \neq \emptyset$

To use KW-container thm, we need to verify that $KG(n, k)$ is locally dense.

For this, we use Expander-mixing lem.

Thm (Alon-Chung): G N -ix D -reg graph.

with min. eigenvalue λ . Then $\forall S \subseteq V(G)$

$$e_G(S) \geq \frac{D}{2N} |S|^2 + \frac{\lambda}{2N} |S| (N - |S|)$$

1st eig. val
↓

error term

Rmk: Main term $\uparrow$ if G is pseudorandom $\lambda \ll D$.

In expectation: $\binom{|S|}{2} \frac{D}{N} = \# \text{ edges}$
 λ tiny

Lovász 1979 The min. eigenvalue of $KG(n, k)$ is

$$\lambda = -\binom{n-k-1}{k-1} = -\frac{k}{n-k} \cdot D, \quad D = \binom{n-k}{k}$$

$$N = \binom{n}{k}$$

Prop (Supersaturation / local denseness)

Given $\varepsilon > 0$, $\forall |S| \geq (1+\varepsilon) \binom{n-1}{k-1}$, $S \subseteq V(KG(n, k))$

$$\Rightarrow e_{KG(n, k)}(S) \geq \left(1 - \frac{1}{1+\varepsilon}\right) \frac{Dn}{N(n-k)} \binom{|S|}{2}$$

pf: Use expander-mixing. β

Rmk Bound on S necessary, as

$$\chi(KG(n, k)) = \binom{n-1}{k-1}$$

Use KW-container to get containers for Kneser graph.

Prop For $\varepsilon > 0$, $2 \leq k \leq \frac{n-1}{2}$, $R = (1+\varepsilon) \binom{n-1}{k-1}$ and

$$g = \frac{1+\varepsilon}{\varepsilon} \cdot \frac{(n-k) \binom{n}{k}}{n \binom{n-k}{k}} \ln \frac{n}{(1+\varepsilon)k}$$

$\Rightarrow \exists$ k -unif $\mathcal{F}_i$ on $[n]$, $1 \leq i \leq \binom{n}{k}$, each of size $\leq R+q$ s.t.

$\forall$ intersecting hyp. $\mathcal{F}$ on $[n]$, $\exists$ some i s.t.
 $\mathcal{F} \subseteq \mathcal{F}_i$

Pf: Apply KW to $KG(n, k)$.

$C_i \subseteq V(KG(n, k)) \rightsquigarrow \mathcal{F}_i$
 $\uparrow$ vcs become edges $\nearrow$ 

Pf (Counting EKR). Verify $e^{-\beta q} N \leq R$

$\Rightarrow \# \text{ int. hyp.} \leq \# \mathcal{F}_i \cdot 2^{R+q} \leq \dots \leq 2^{(1+\epsilon k) \binom{n-1}{k-1}}$ 

Rank Bound $n \geq 2k+1$ is optimal.

$$n=2k, \quad \binom{n-1}{k-1} = \frac{1}{2} \binom{n}{k}$$

$(A, \bar{A})$ pair $\begin{cases} - \emptyset \\ - A \\ - \bar{A} \end{cases} \Rightarrow 3^{\frac{1}{2} \binom{n}{k}} = 3^{\binom{n-1}{k-1}}$

